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Design of Acoustic Liner in Small Gas Turbine Combustor Using One-Dimensional Impedance Models

Daesik Kim[1]
School of Mechanical and Automotive Engineering,
Gangneung-Wonju National University,
150 Namwon-ro, Wonju 26403,
Gangwon, South Korea
e-mail: dkim@gwnu.ac.kr

Seungchai Jung
Gas Turbine Development Team,
Hanwha Aerospace R&D Center,
471 Pangyo, Bundang, Seongnam 13521,
Gyeonggi, South Korea

Heeho Park
Gas Turbine Development Team,
Hanwha Aerospace R&D Center,
471 Pangyo, Bundang, Seongnam 13521,
Gyeonggi, South Korea

The side-wall cooling liner in a gas turbine combustor serves main purposes—heat transfer and emission control. Additionally, it functions as a passive damper to attenuate thermoacoustic instabilities. The perforations in the liner mainly convert acoustic energy into kinetic energy through vortex shedding at the orifice rims. In the previous decades, several analytical and semi-empirical models have been proposed to predict the acoustic damping of the perforated liner. In the current study, a few of the models are considered to embody the transfer matrix method (TMM) for analyzing the acoustic dissipation in a concentric tube resonator with a perforated element and validated against experimental data in the literature. All models are shown to quantitatively appropriately predict the acoustic behavior under high bias flow velocity conditions. Then, the models are applied to maximize the damping performance in a realistic gas turbine combustor, which is under development. It is found that the ratio of the bias flow Mach number to the porosity can be used as a design guideline in choosing the optimal combination of the number and diameter of perforations in terms of acoustic damping. [DOI: 10.1115/1.4040765]

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Introduction

In the areas of modern aero-engines and industrial gas turbines, it is a necessity not only to maintain high performance but also to cut down fuel consumption and reduce pollutant emissions. One of the promising solutions for achieving these requirements is to adapt a lean premixed combustion technology. However, this type of combustion is intrinsically susceptible to a combustion instability, which is a coupling mechanism between acoustic pressure fluctuations and unsteady heat release oscillations. Consequently, this can lead to a crucial damage to the combustor and turbine parts, and reduce the system efficiency and life of the engine components, such as rotor blades [1].

To minimize damages resulting from the combustion instability in a gas turbine combustor, one typical approach is to optimize the acoustic absorption capabilities of the perforated combustor liner. It is well known that perforations can generally be applied to damp acoustic perturbations. The original functions of the liner are to contain the combustion process and to facilitate the distribution of air to all various combustor zones. Of the several combustors currently in service, up to 20% of the total combustor air flow is employed in liner wall cooling [2]. As previously mentioned, another particular feature of the liners is that they are proven to have a significant effect on the damping mechanisms of the combustor acoustics. The liner damping mainly involves vortices generated over the rims of the perforations. When incident sound waves interact with and destabilize shear layers formed at the orifice rims, vortex rings are formed and the sound energy is converted into kinetic energy, which is dissipated by the surrounding fluid [3–7].

A number of research groups have reported their results on the damping effect of cooling liners or orifices, because they were proven to be effective in suppressing combustion instabilities in rocket engines in the 1960s [8,9]. Most recently, Zhao and Li [5], Lahiri and Bake [10], and Lahiri [11] presented excellent reviews summarizing the status, challenges, and progress of implementing acoustic dampers, such as perforated liners on various systems. According to their reviews, most acoustic liner designs in the practical field are conducted using analytical or semi-empirical approaches because the holes involved are extremely small to accurately simulate them with most computational codes (Typically, the hole diameters range from 0.64 to 8.82 mm [12].).

There exists a few analytical or semi-empirical modeling approaches to calculate the damping characteristics in perforations. The least cumbersome and simplest analytic modeling method is to use a zero-dimensional approach [5,6] where the acoustic impedance terms are educed from measurements of multiple liner geometries. Since Howe [4,13] presented the first model for a perforated plate with bias flow based on Crandall's acoustic theory [14], there have been comprehensive impedance model developments. His model is regarded as classical standards for modeling bias flow orifices or perforations. Jing and Sun [15,16] and Bellucci et al. [17,18] presented an extended and/or modified Howe model and conducted various parametric studies on perforated plates with bias flow. Similarly, combining various empirical correlations and theoretical findings for key acoustic parameters in a perforated liner, Elnady and Bodén [6], Schulz et al. [19], and Kraft et al. [20] extended the models of Crandall and Howe.

However, these zero-dimensional models cannot globally consider the effect of the acoustic field in a duct where traveling or standing waves exist. Especially, in a gas turbine combustor, acoustic damping characteristics of liners are determined by various interactive influences of acoustic fields in the combustor (or duct), annulus (or cavity), and in-between perforations. These complex interactions between duct acoustics and perforation impedance can generally be modeled by using the transfer matrix method (TMM), which was formulated from the electrical circuit theory [21]. In this method, acoustic fields in the duct and cavity are described by their respective differential equations. Thereafter, they are combined by several small control volumes.

The objective of the current study is to briefly summarize various zero-dimensional impedance models for perforated liners with bias flow. Subsequently, the selected impedance models are applied to the one-dimensional (1D) TMM model to evaluate the acoustic damping performance of side-wall liners. To validate our liner acoustic model, Lahiri's [10,11] measurement results are

[1]Corresponding author.

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chosen because they provide a considerable amount of parametric measurement results with various geometries of a perforation liner. To the authors' best knowledge, most existing publications on the zero- or one-dimensional analytical studies on the perforated liner are limited to measurements in impedance tubes or lab-scale combustors. The actual operating conditions of gas turbine liners such as flow rate, pressure, and temperature cannot be exactly implemented in a lab-scale measurement; additionally, publications with detailed information of the combustor configurations and flow conditions are often restricted for security reasons during the development stages. Accordingly, the available investigations on the damping performance in a real gas turbine combustor have been rare hitherto. In the present study, the selected impedance models in a TMM framework are applied to the geometry and operating conditions of a practical acoustic liner. Subsequently, we compare the acoustic absorption characteristics calculated using each model, and finally, provide the quantitative analysis results on the effects of key design parameters on the liner acoustic damping in the geometry of a small gas turbine combustor.

Modeling Approach

Definition of Perforation and Acoustic Damping Parameters. Figure 1 shows the main parameters affecting acoustic impedance in perforated liners with bias flow. The combustor flow (commonly called "grazing flow" in the acoustic field with perforation plates) is determined as the main flow from the combustor dome inlet to the exit (or turbine inlet) and parallel to the liner face. On the other hand, the bias flow is the introduction of airflow perpendicular to the perforations of the liner and through the holes. The cavity is determined as the space between the outer casing and the liner. The porosity (σ) is defined as the ratio of the open area (created by each perforation of uniform diameter (d) and the number of holes (n)) to the liner surface area (A_{surf}), i.e., $\sigma = \pi n d^2/4A_{\text{surf}}$.

The normalized acoustic impedance (z) is defined as the ratio of the complex acoustic pressure ($\hat{p}$) to the normal component of the complex acoustic velocity ($\hat{u}$)

$$z = \frac{1}{\rho c}\frac{\hat{p}}{\hat{u}} = \Theta + iX \tag{1}$$

where ρ and c are the flow density and speed of sound, respectively. The real part (Θ) is referred to as the resistance and contributes to the dissipative processes occurring in liner orifices. The imaginary part (X) is called reactance and represents the inertia of the fluid fluctuating into the orifices and in the backing space under the effect of the acoustic field [17,21].

Impedance Model of Perforations. There are numerous impedance models suggested for perforations with bias flow. Detailed reviews on various types of models can be found in a few literature [5,6,10]. In this paper, the four most widely used impedance models are considered as follows.

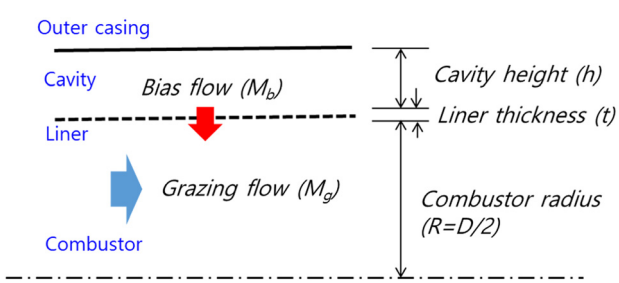

Fig. 1 Definition of key parameters in an acoustic perforated liner

Jing Analytical Model. The model of Jing is developed from the Howe model, which analytically derives a formulation of the Rayleigh conductivity (K_R) for a circular aperture in an infinitely thin plate subject to low Mach number bias flow. Howe [4,13] suggested the following expression for the Rayleigh conductivity for low Mach number bias flow through the perforations:

$$K_R = (\gamma + i\delta)d \tag{2}$$

$$\gamma + i\delta = 1 + \frac{\frac{\pi}{2}I_1(\text{St})e^{-\text{St}} + iK_1(\text{St})\sinh(\text{St})}{\text{St}\left[\frac{\pi}{2}I_1(\text{St})e^{-\text{St}} - iK_1(\text{St})\cosh(\text{St})\right]} \tag{3}$$

Here, St $(= \omega d/(2u_b))$ is the Strouhal number based on the bias flow velocity (u_b), ω is the angular frequency, and I_1 and K_1 are the modified Bessel functions of the first and second kind, respectively. Jing and Sun [15,16] extended the Howe model by including the wall thickness (t) and recasting the Rayleigh conductivity in Howe's expression in terms of the normalized acoustic impedance as follows:

$$z_{\text{Jing}} = \frac{k\pi d\delta}{4(\gamma^2+\delta^2)} + i\left(\frac{k\pi d\gamma}{4(\gamma^2+\delta^2)} + kt\right) \tag{4}$$

where k $(= \omega/c)$ is the wavenumber.

The Jing model is also called a modified Howe model by a number of succeeding research. It has been validated by several experimental investigations [22–27] and exhibits a general agreement with measured data.

Betts Analytical Model. Betts [28,29] modified the classical Crandall [14] and Melling [30] impedance models to account for the high Mach number bias flow and to obtain an intuitive impedance formulation without the Bessel functions. This was achieved by combining low and high shear number approximations and adding the Helmholtz approximation, so that

$$z_{\text{Betts}} = \frac{16\nu t}{c\sigma C_d d^2} + \frac{\sqrt{2\omega\nu}}{c\sigma C_d}\frac{2t}{d} + \frac{(1-\sigma^2)}{c\sigma C_d}\left(\frac{\sqrt{2}p_{\text{ref}}10^{\text{SPL}/20}}{2\rho c^2 C_d} + u_b\right) + i\left(\frac{kt}{cC_d} + \frac{\sqrt{2\omega\nu}}{c\sigma C_d}\frac{2t}{d} + \frac{8d}{3\pi\psi}\right) \tag{5}$$

where ν is the kinematic viscosity; C_d is the orifice discharge coefficient; the reference pressure (p_{ref}) is 2×10^5 Pa; SPL represents the sound pressure level, in dB; and $\psi(= 1 - \sqrt{0.5\sigma})$ is the interaction factor for adjacent orifices [30].

Bauer Model. Bauer [31] proposed an impedance model combining several of the previous empirical and analytical correlations, such as a viscous term from Ingard and Ising [32], bias flow term from Zinn [33], terms for a grazing flow, and a reactance from the NASA technical reports [34,35]

$$z_{\text{Bauer}} = \frac{\sqrt{8\nu\omega}}{\rho c\sigma}\left(1 + \frac{t}{d}\right) + \frac{0.3M_g}{\sigma} + \frac{4}{3\pi C_d^2}\frac{M_b}{\sigma} + i\frac{k(t+0.25d)}{\sigma} \tag{6}$$

Equation (6) is referenced by several literature [5,23,36,37] as a benchmark case in developing an empirical impedance model.

Elnady Model. Elnady and Bodén [6] extended the Bauer approach through further literature reviews and comprehensive experimental parametric study. They unified previous analytical and empirical formulae, such as the internal loss term from Crandall [14], nonlinear and radiation terms from Melling [30] and Ingard and Ising [32], and bias and grazing flow terms from Bauer [31]. However, a number of these parameters are modified through validation measurement and analytical derivation. The

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proposed model can be summarized in the following equations for the perforated liner resistance and reactance:

$$z_{\text{Elnady}} = \Theta_{\text{Elnady}} + iX_{\text{Elnady}} \tag{7}$$

where

$$\Theta_{\text{Elnady}} = \text{Re}\left\{\frac{ik}{\sigma C_d}\left[\frac{t}{F(K_s' d/2)} + \frac{t_{\text{end_re}}}{F(K_s d/2)}\psi\right]\right\} + \frac{1-\sigma^2}{\sigma^2 C_d^2}\frac{|\hat{u}|}{2c} + \frac{1}{\sigma}\left[1 - \frac{2J_1(kd)}{kd}\right] + \frac{0.5M_g}{\sigma} + \frac{1.15M_b}{\sigma C_d} \tag{8}$$

$$X_{\text{Elnady}} = \text{Im}\left\{\frac{ik}{\sigma C_d}\left[\frac{t}{F(K_s' d/2)} + \frac{t_{\text{end_im}}}{F(K_s d/2)}\psi\right]\right\} - \frac{1-\sigma^2}{\sigma^2 C_d^2}\frac{|\hat{u}|}{6c} + \frac{1}{\sigma}\left[1 - \frac{2J_1(kd)}{kd}\right] - \frac{0.3M_g}{\sigma} - \cot(kh) \tag{9}$$

where $F(x) = 1 - 2J_1(x)/x/J_0(x)$; J_0 and J_1 are the zeroth- and first-order Bessel function of the first kind, respectively; the viscous Stokes wavenumber, without considering any thermal conductivity effects, is $K_s = \sqrt{-i\omega/\nu}$; K_s' is the effective Stokes wavenumber considering the viscosity and thermal conductivity losses near a highly conductive wall and is determined as $K_s' = \sqrt{-i\omega/\nu'}$. Here, ν' represents the effective kinematic viscosity, including heat conduction effects. $|\hat{u}|$ is the amplitude of the normal acoustic velocity, which can be determined iteratively from $|\hat{u}| = \text{SPL}/(\rho c\sqrt{\Theta^2 + X^2})$. As seen in the equation, the end correction parameter is included in both resistance ($t_{\text{end,re}}$) and reactance ($t_{\text{end,im}}$) terms. Zhao and Li [5] suggested $t_{\text{end,re}} = 0.2d + 200d^2 + 16000d^3$ and $t_{\text{end,im}} = 0.5d$ from the curve (or linear)-fit of the measured data. On the other hand, Betts, in Eq. (5), only used it in the reactance term of $t_{\text{end,im}} = 8d/3\pi \approx 0.85d$ but did not consider its effect on the resistance. Moreover, the grazing and bias flow coefficients in the resistance are replaced with 0.5 and $1.15C_d$, from 0.3 and $4/(3\pi C_d^2)$, respectively, in the Bauer model through their own parametric test results. Furthermore, the grazing flow and cavity effects are added to the reactance term.

Transfer Matrix Method for One-Dimensional Acoustic Analysis. The plane wave reflection and absorption coefficients of a perforated liner with a normal sound incidence can be directly obtained when its impedance is given. This is the situation under which the aforementioned impedance models are validated. A typical example of an acoustic experiment setup in an impedance tube with a perforated plate is shown in Fig. 2.

However, when the same liner with an unpartitioned cavity is mounted as a side-wall with a grazing sound incidence, such as in a gas turbine combustor liner, the sound propagation within the cavity is required to be taken into account. For this type of liner, the acoustics inside the duct and cavity are coupled through the impedance boundary of the perforated liner in-between them. These complex interactions between duct acoustics and perforation impedance can be generally modeled by using TMM, which is a convenient approach for modeling one-dimensional, linear dynamic systems and is well described in several textbooks [21,37] and papers [10,11]. In this paper, the general description is summarized as follows.

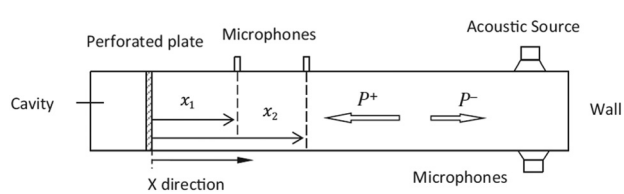

Fig. 2 Typical example of acoustic damping measurement setup for a perforated plate with normal incidence wave [26]

Figure 3 shows the setup describing a generalized TMM approach in a two-duct system with a perforated element. From the mass continuity and momentum balance equations in the common section of Fig. 3(*a*), acoustic pressures and velocities at $x = 0$ can be related to those at $x = l$ through the transfer matrix relationship

$$\begin{bmatrix} \hat{p}_1(0) \\ \hat{p}_2(0) \\ \rho c\hat{u}_1(0) \\ \rho c\hat{u}_2(0) \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} & T_{13} & T_{14} \\ T_{21} & T_{22} & T_{23} & T_{24} \\ T_{31} & T_{32} & T_{33} & T_{34} \\ T_{41} & T_{42} & T_{43} & T_{44} \end{bmatrix} \begin{bmatrix} \hat{p}_1(l) \\ \hat{p}_2(l) \\ \rho c\hat{u}_1(l) \\ \rho c\hat{u}_2(l) \end{bmatrix} \tag{10}$$

where $[\mathbf{T}] = [\mathbf{A}(0)][\mathbf{A}(l)]^{-1}$. Elements of the matrix $[\mathbf{A}(x)]$ are given by $i = 1, 2, 3, 4$

$$A_{1,i} = \Psi_{3,i}e^{\beta_i x},\ A_{2,i} = \Psi_{4,i}e^{\beta_i x},\ A_{3,i} = -\frac{\Psi_{1,i}e^{\beta_i x}}{ik_0 + M_1\beta_i},\quad A_{4,i} = -\frac{\Psi_{2,i}e^{\beta_i x}}{ik_0 + M_2\beta_i} \tag{11}$$

$[\boldsymbol{\Psi}]$ and $\{\boldsymbol{\beta}\}$ are, respectively, the eigenmatrix and the eigenvector of the matrix

$$\begin{bmatrix} -\alpha_1 & -\alpha_3 & -\alpha_2 & -\alpha_4 \\ -\alpha_5 & -\alpha_7 & -\alpha_6 & -\alpha_8 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{12}$$

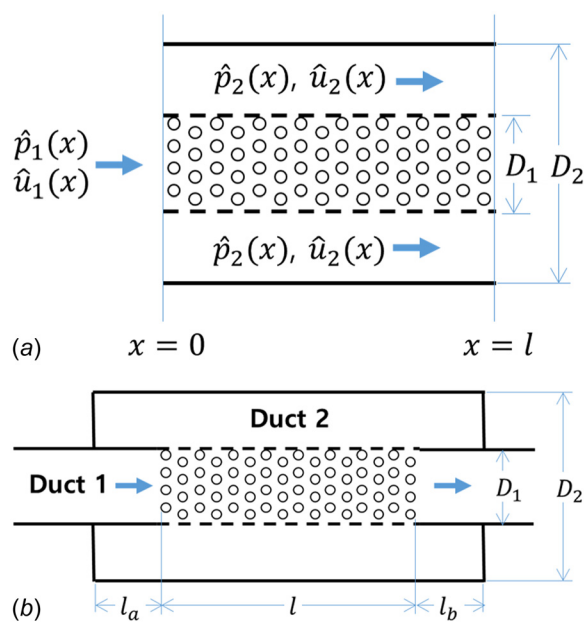

Fig. 3 Typical example of an acoustic damping measurement setup for a perforated plate with normal incidence wave (modified from Refs. [21,37]): (*a*) common two-duct perforated section and (*b*) concentric tube resonator

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where

$$\alpha_1 = -\frac{iM_1}{1-M_1^2}\left(\frac{k_a^2+k_0^2}{k_0}\right), \quad \alpha_2 = -\frac{k_a^2}{1-M_1^2},$$
$$\alpha_3 = \frac{iM_1}{1-M_1^2}\left(\frac{k_a^2-k_0^2}{k_0}\right), \quad \alpha_4 = -\left(\frac{k_a^2-k_0^2}{1-M_1^2}\right)$$
$$\alpha_5 = \frac{iM_2}{1-M_2^2}\left(\frac{k_b^2-k_0^2}{k_0}\right), \quad \alpha_6 = -\left(\frac{k_b^2-k_0^2}{1-M_2^2}\right),$$
$$\alpha_7 = -\frac{iM_2}{1-M_2^2}\left(\frac{k_b^2+k_0^2}{k_0}\right), \quad \alpha_8 = \frac{k_b^2}{1-M_2^2} \tag{13}$$

with

$$k_0 = \frac{\omega}{c}, \quad k_a^2 = k_0^2 - \frac{4ik_0}{D_1 z}, \quad k_b^2 = k_0^2 - \frac{4ik_0 D_1}{(D_2^2 - D_1^2)z} \tag{14}$$

Here, the normalized specific impedance (z) of the perforated element is obtained from Eqs. (4) to (7). However, in the case of the Elnady model in Eq. (9), the cavity term "$-\cot(kh)$" is eliminated when combined with TMM. In Fig. 3, D_1 and D_2 are the duct diameters, and M_1 and M_2 are the Mach numbers at each duct, respectively.

The solid wall boundary conditions at both ends of the cavity in Fig. 3(*b*) can be represented by [21]

$$z_2(0) = -\frac{\hat{p}_2(0)}{\rho c \hat{u}_2(0)} = -i\cot(k_0 l_a) \tag{15}$$

$$z_2(l) = \frac{\hat{p}_2(l)}{\rho c \hat{u}_2(l)} = i\cot(k_0 l_b) \tag{16}$$

The transfer matrix for the setup of the concentric tube resonator in Fig. 3(*b*) can be obtained from the 4×4 matrix $[T]$ by applying Eqs. (15) and (16)

$$\begin{bmatrix} \hat{p}_1(0) \\ \rho c \hat{u}_1(0) \end{bmatrix} = \begin{bmatrix} T_a & T_b \\ T_c & T_d \end{bmatrix} \begin{bmatrix} \hat{p}_1(l) \\ \rho c \hat{u}_1(l) \end{bmatrix} \tag{17}$$

where

$$T_a = T_{11} + A_1 A_2, T_b = T_{13} + B_1 A_2, T_c = T_{31} + A_1 B_2, T_d = T_{33} + B_1 B_2 \tag{18}$$

with

$$A_1 = \frac{X_1 T_{21} - T_{41}}{F_1}, \quad B_1 = \frac{X_1 T_{23} - T_{43}}{F_1}, \quad A_2 = T_{12} + X_2 T_{14},$$
$$B_2 = T_{32} + X_2 T_{34}$$
$$X_1 = -i\tan(k_0 l_a), \quad X_2 = i\tan(k_0 l_b),$$
$$F_1 = T_{42} + X_2 T_{44} - X_1(T_{22} + X_2 T_{24}) \tag{19}$$

Damping Performance Parameters. In order to evaluate the acoustic damping performance of the liner and to validate the modeling results against the measurements in the benchmark works [10,11], the energy dissipation coefficient (DC) is defined in the same manner as several literature do. The summation of the reflection, transmission, and dissipation coefficients is unity, as follows:

$$\mathrm{RC}^{\pm} + \mathrm{TC}^{\pm} + \mathrm{DC}^{\pm} = 1 \tag{20}$$

where RC and TC represent the energy reflection and transmission at the perforated liner, respectively; the superscripts, "+" and "−" are the positive and negative wave propagation in the x-direction, respectively. Then, RC and TC at each direction of the perforated liner are defined as (when end-reflections are neglected)

$$\mathrm{RC}^+ = \frac{\hat{p}_{\mathrm{up}}^-}{\hat{p}_{\mathrm{up}}^+}, \ \mathrm{RC}^- = \frac{\hat{p}_{\mathrm{down}}^+}{\hat{p}_{\mathrm{down}}^-}, \ \mathrm{TC}^+ = \frac{\hat{p}_{\mathrm{down}}^+}{\hat{p}_{\mathrm{up}}^+}, \ \mathrm{TC}^- = \frac{\hat{p}_{\mathrm{up}}^-}{\hat{p}_{\mathrm{down}}^-} \tag{21}$$

where the subscripts "up" and "down" represent the upstream and downstream of perforated liner (Duct 1 in Fig. 3(*b*)), respectively. By combining Eq. (20) with Eq. (21), the positive-direction, negative-direction, and their average dissipation coefficients can be represented by

Dissipation coefficient in the positive direction (DC^+)

$$\mathrm{DC}^+ = 1 - \left[\frac{(1-M_{\mathrm{up}})^2}{(1+M_{\mathrm{up}})^2}\cdot\left|\mathrm{RC}^+\right|^2 + \frac{(1+M_{\mathrm{down}})^2}{(1+M_{\mathrm{up}})^2}\cdot\left|TC^+\right|^2\right]$$
$$= 1 - \left[\frac{(1-M_{\mathrm{up}})^2\cdot\left|\hat{p}_{\mathrm{up}}^-\right|^2 + (1+M_{\mathrm{down}})^2\cdot\left|\hat{p}_{\mathrm{down}}^+\right|^2}{(1+M_{\mathrm{up}})^2\cdot\left|\hat{p}_{\mathrm{up}}^+\right|^2}\right] \tag{22}$$

Dissipation coefficient in the negative direction (DC^-)

$$\mathrm{DC}^- = 1 - \left[\frac{(1+M_{\mathrm{up}})^2}{(1-M_{\mathrm{up}})^2}\cdot\left|\mathrm{RC}^-\right|^2 + \frac{(1-M_{\mathrm{down}})^2}{(1-M_{\mathrm{up}})^2}\cdot\left|\mathrm{TC}^-\right|^2\right]$$
$$= 1 - \left[\frac{(1+M_{\mathrm{up}})^2\cdot\left|\hat{p}_{\mathrm{down}}^+\right|^2 + (1-M_{\mathrm{down}})^2\cdot\left|\hat{p}_{\mathrm{up}}^-\right|^2}{(1-M_{\mathrm{up}})^2\cdot\left|\hat{p}_{\mathrm{down}}^-\right|^2}\right] \tag{23}$$

Average dissipation coefficient ($\overline{\mathrm{DC}}$)

$$\overline{\mathrm{DC}} = \frac{\mathrm{DC}^+ + \mathrm{DC}^-}{2} \tag{24}$$

The detailed information on the derivation process of the DC is well described in Lahiri's works [10,11]. The symmetric liner configuration and zero grazing flowrate ($M_g = 0$) lead to $\mathrm{DC}^+ = \mathrm{DC}^- = \overline{\mathrm{DC}}$. However, an asymmetric liner configuration or a grazing flow yields $\mathrm{DC}^+ \neq \mathrm{DC}^-$. In that case, $\overline{\mathrm{DC}}$ allows the evaluation of the overall performance of the liner, considering both directions.

Another parameter used in this paper for evaluating the damping performance of a particular combustor liner is the dissipation integral (DI) defined in Eq. (25) [37]. It gives the average value of the DC over the frequency (f) range under consideration. Depending on the values chosen for the minimum ($f_{\min}$) and maximum ($f_{\max}$) frequencies of interest, the dissipation integral defines a relatively broader or narrower damping.

Dissipation integral

$$\mathrm{DI} = \frac{1}{(f_{\max} - f_{\min})}\int_{f_{\min}}^{f_{\max}} \overline{DC}\, df \tag{25}$$

Model Validation

Selected Validation Cases. In order to suggest an optimal acoustic impedance model for a perforated liner with a bias flow and to validate the suggested model, Lahiri's measurement results [10,11] are chosen in the current study. For characterizing the acoustic impedance of perforated liners, impedance measurement tests were conducted over a frequency range of 200–1000 Hz by using two external loudspeakers installed at both ends of their own "Duct acoustic test rig," as shown in Fig. 4(*a*). Two types of liners in their experiment are selected for the model validation as

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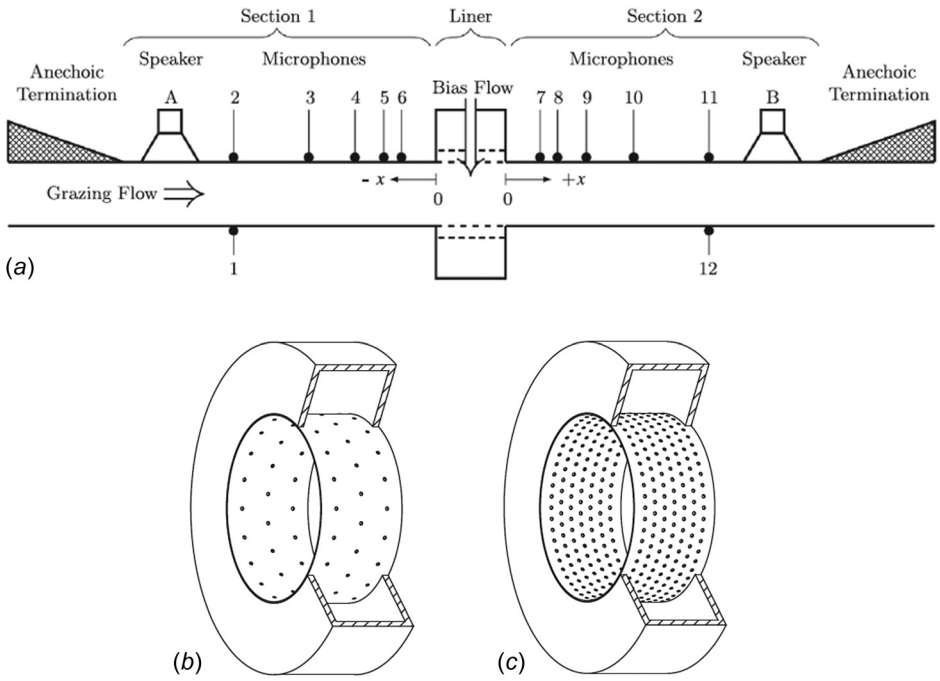

Fig. 4 Benchmark duct acoustic test rig and selected perforated liner geometry [10,11]: (*a*) duct acoustic test rig, (*b*) liner A (low perforation), and (*c*) liner B (high perforation)

illustrated in Figs. 4(*b*) and 4(*c*). The two liners basically have the same geometries, but different porosities. The specific interest of the current study is the high bias flow condition, such as that in the operating circumstance of a practical gas turbine. Under such a high bias flow condition, it is generally reported that the key parameters affecting the damping performance of perforated liners are the bias flow Mach number (M_b), grazing flow Mach number (M_g), and porosity (σ) [10,11,24,26,27]. Therefore, the other parameters are fixed, as listed in Table 1. A more detailed information on the test setup, conditions, and other parametric study experimental results can be found in the literature.

Validation Results. Figure 5 shows the validation results against benchmark measurements for different bias flow Mach numbers and porosities. Lahiri and Bake [10,11] have shown that the DC without bias flow ($M_b = 0$), which is not presented here, exhibits a sharp peak approximating the resonance frequency of the cavity. However, with increasing M_b at a fixed porosity, the level of dissipation is first increased and then reduced, while the dissipation further becomes a broadband. In the configuration of liner A, the optimum M_b to achieve a good broadband damping is measured to be 0.049 [10]. As shown in Fig. 5(*a*), the increase in

Table 1 Specifications of the studied liner configurations and operating conditions

	Liner A	Liner B
Liner diameter, D_1 (mm)	140	140
Liner length, l (mm)	60	60
Orifice diameter, d (mm)	2.5	2.5
Orifice number, n (ea)	60	364
Porosity, σ (%)	1.0	6.8
Liner thickness, t (mm)	1	1
Cavity height, h (mm)	50	50
Temperature, $\overline{T}$ (K)	293	293
Static pressure, $\overline{p}$ (bar)	1	1
SPL (dB)	100	100
Discharge coefficient of orifice, C_d	0.64	0.52

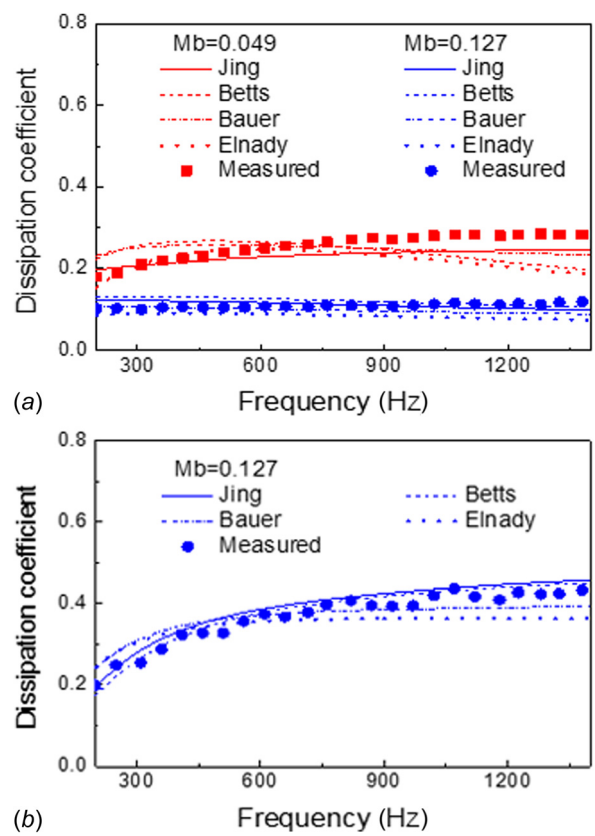

Fig. 5 Model validation results for bias flow Mach number effect ($M_g = 0$). Models of Jing, Betts, Bauer, and Elnady are from Eqs. (4), (5), (6), and (7)–(9), respectively, and measurement data are from Ref. [10]. (*a*) Liner A (low porosity) and (*b*) liner B (high porosity).

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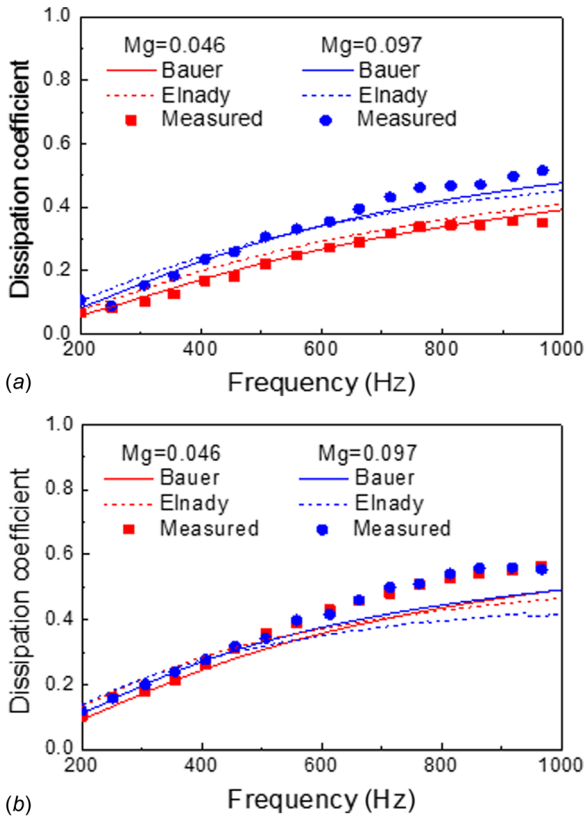

Fig. 6 Model validation results for simultaneous bias and grazing flow conditions. Models of Bauer and Elnady are from Eqs. (6) and (7)–(9), respectively, and measurement data are from Ref. [10]: (*a*) $M_b = 0.016$ and (*b*) $M_b = 0.035$.

M_b from 0.049 to 0.127 leads to lower dissipation characteristics of liner A ($\sigma = 1.0\%$), and the damping of $M_b = 0.127$ becomes extremely inadequate. However, at the same value of $M_b = 0.127$, the high porosity liner (liner B ($\sigma = 6.8\%$)) exhibits a considerably improved acoustic damping, which means that the perforated liner dissipation is greatly affected not only by the porosity but also by the bias flow velocity.

Four different impedance models, explained in the Modeling Approach section, were tested and validated against the measurements. Apparently, there is no perfect model that can quantitatively match the experimental data. However, all models can qualitatively well represent the measured damping trend relating to both M_b and σ. A general agreement between the experiment and the prediction for all models was also found under other bias flow conditions in the two liner configurations. However, for the current validation, the optimal impedance model among the models of Jing, Betts, Bauer, and Elnady could not simply be proposed based on tested results.

Figure 6 shows the validation results of the grazing flow Mach number (M_g) effect at two different bias flow velocity conditions in liner B. Only two models of Bauer and Elnady, where the grazing flow effect is considered in the models, are included in the plot. For a slow bias flow condition, as in Fig. 6(*a*), the increase in M_g shows a positive effect on the acoustic absorption of a perforated liner and is well captured in both models. Meanwhile, for a higher M_b of 0.035 in Fig. 6(*b*), no remarkable M_g effect is found. Lahiri and Bake [10] reported, with reference to the previous literature [35,36], that the influence of grazing flow on the acoustic absorption is similar to the bias flow effect. However, the acoustic dissipation is independent of the grazing flow when $M_b/M_g > 0.3$. From the plots, we found that both models quantitatively yield a similar M_g effect on the dissipation, which is in good agreement with the experimental data.

Acoustic Liner Design for a Small Gas Turbine Combustor

Target Combustor and Design Conditions. Hanwha Aerospace Co., Ltd. (Seongnam, South Korea) is developing a small size of lean premixed combustor applicable for a 25-kN aircraft engine. The combustor is composed of 12 swirlers and fuel nozzle assemblies set in a circumferential direction. A section view of the combustor under development is shown in Fig. 7(*a*), and its schematic for the acoustic characterization of the liner wall cooling is shown in Fig. 7(*b*). The combustor height is averaged over the axial direction in Fig. 7(*a*) and taken as D_1 in Fig. 7(*b*). The cavity height (h) is calculated from the average values of the inner and outer annulus heights. Tables 2 and 3, respectively, summarize the geometrical parameters and operating conditions at a design point from the engine cycle evaluation processes. The current study is focused on determining the optimal liner perforation parameters, such as bias flow Mach number (M_b) and porosity (σ), which are the key factors affecting the liner acoustic damping under high bias flowrate conditions. Once the liner cooling flowrate (total bias flowrate, $\dot{m}_b$, not flowrate per hole), combustor length (l_{tot}), and diameter (D_1) are provided, both M_b and σ are determined by the number and diameter of liner holes, as shown in the following equations:

$$M_b = \frac{V_b}{c} = \frac{1}{c}\left[\frac{\dot{m}_b}{n}\cdot\frac{4}{\pi\rho d^2}\right] = \frac{4\dot{m}_b}{\pi\rho c n d^2} \tag{26}$$

$$\sigma = \frac{A_{\text{open}}}{A_{\text{surf}}} = \frac{\pi n d^2/4}{\pi D_1 l_{\text{tot}}} = \frac{n d^2}{4 D_1 l_{\text{tot}}} \tag{27}$$

Acoustic Damping Evaluation of the Liner. The amplitude of the acoustic pressure oscillations in a combustor has a significant effect on the acoustic damping characteristics of a perforated liner

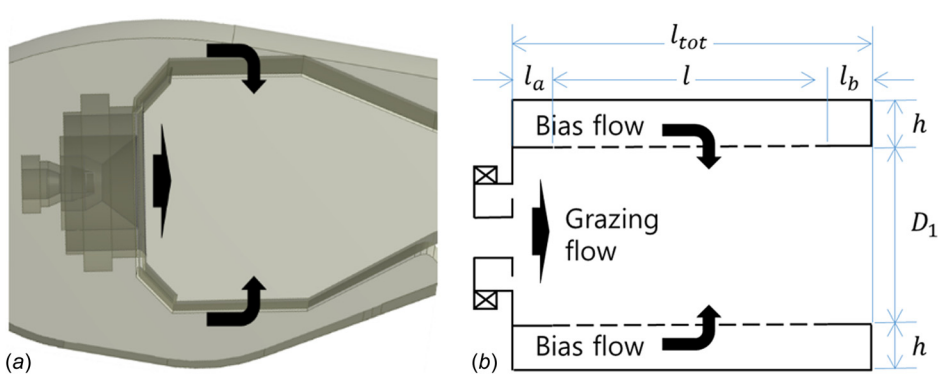

Fig. 7 Practical gas turbine combustor under consideration: (*a*) sectional view and (*b*) schematic diagram

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Table 2 Specifications of geometrical parameters of practical gas turbine combustor

Combustor diameter, D_1 (mm)	110
Lengths, l_a, l, l_b, l_{tot} (mm)	0, 140, 15, 155
Cavity depth, h (mm)	16
Liner thickness, t (mm)	1.2

Table 3 Gas properties for acoustic liner modeling

	Bias flow	Grazing flow
Pressure, $\bar{p}$ (bar)	19.5	19.5
Temperature, $\bar{T}$ (K)	732	1600
Flowrate, $\dot{m}$ (kg/s)	5.8	16.1
Speed of sound, c (m/s)	539.6	778.5
Density, ρ (kg/m^3)	9.2	4.2
Mach number, M_b or M_g	0.03–0.21	0.06

[5,6,11,17,28]. The influence of the SPL on the dissipation is shown in Fig. 8 for two different frequencies of 700 Hz and 1700 Hz using two impedance models, i.e., of Betts and Elnady. In the figures, the porosity is fixed at 0.02 and the selected bias flow Mach numbers are 0.03 and 0.1, in Figs. 8(*a*) and 8(*b*), respectively. At each M_b, the SPL is varied in the range of 80–150 dB and other properties are specified in Tables 2 and 3. Here, the dissipation is normalized by the value at the lowest SPL (80 dB in the current study) to remove the distraction from different dissipation levels by different frequencies and bias flow velocities, and to focus on the nonlinear behavior.

The prediction results using the Elnady model show that all of the tested SPL remain under the linear regime, which means that the dissipation level is maintained constant over the pressure amplitudes. However, the calculation results using the Betts model reveal a strong dependency on the SPL. This is primarily due to the fact that the nonlinear term in the Betts impedance model of Eq. (5) is an exponential function of SPL, while that in the Elnady model of Eqs. (8) and (9) is linearly dependent upon the pressure amplitude. In addition, we found in the dissipation results of the Betts model that the transition between linear and nonlinear occurs at a lower SPL in the slower bias flow condition of Fig. 8(*a*), compared with the greater M_b case of Fig. 8(*b*). This behavior agrees with the experimental findings of Rupp and Carrotte [38], Lahiri [11], and Tran et al. [39]. Following the current results, the SPL for other parametric studies in this work is fixed at 100 dB, where the acoustic damping characteristics show a linear behavior for all of the considered M_b and frequencies.

Figure 9 shows the DC calculation results of the acoustic liner under consideration as listed in Table 2, using four different impedance models for a given porosity of 0.02 and two different bias flow Mach numbers of 0.1 and 0.18. Both M_b cases show the broadband damping feature over a tested frequency, which is identified as being typical under such a high bias flow velocity condition [10]. By comparing Figs. 9(*a*) and 9(*b*), the increase in M_b consequently results in the reduction of the DC for a given porosity and operating condition. All models show similar behaviors although the Elnady model is found to predict a slightly lower DC than the other three models do. Figure 10 compares the normalized resistance and reactance of each model as a function of the frequency under the same condition as Fig. 9(*b*). In all models, the resistance remains

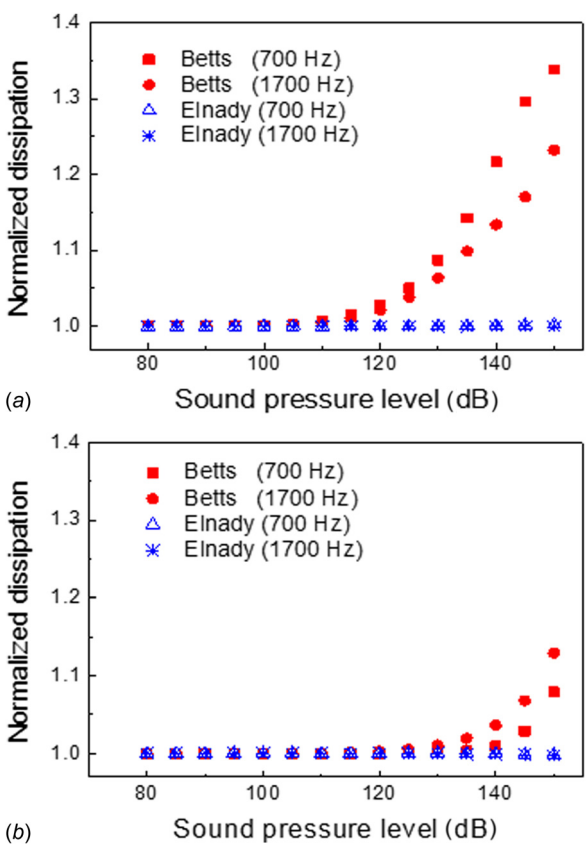

Fig. 8 Normalized dissipation as a function of SPL ($M_g = 0.06$ and $\sigma = 0.02$). The dissipation is normalized by the value at the SPL of 80 dB for each condition. Models of Betts and Elnady are from Eqs. (5) and (7)–(9), respectively. (*a*) $M_b = 0.03$ and (*b*) $M_b = 0.1$.

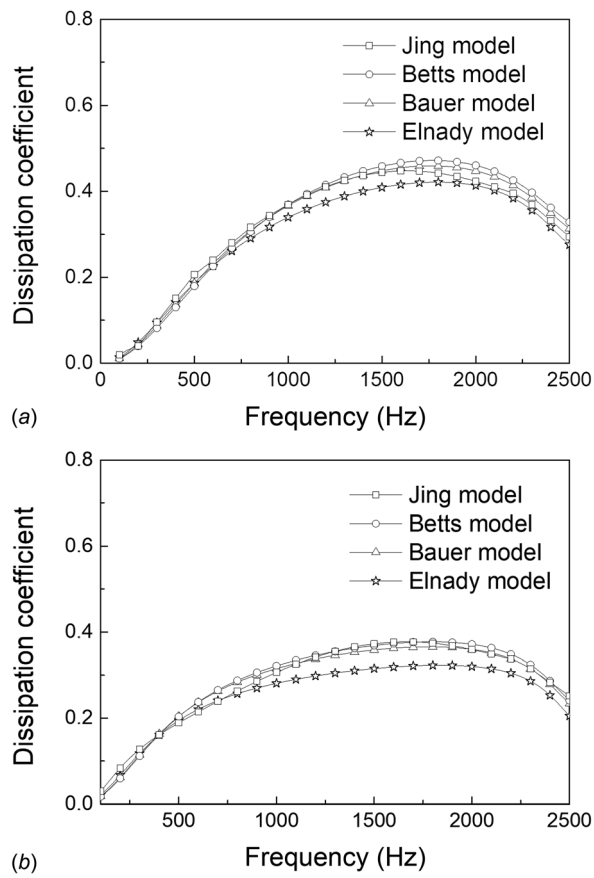

Fig. 9 Dissipation coefficient calculation results as a function of frequency according to impedance models ($M_g = 0.06$, $\sigma = 0.02$, and SPL = 100 dB). Models of Jing, Betts, Bauer, and Elnady are from Eqs. (4), (5), (6), and (7)–(9), respectively: (*a*) $M_b = 0.1$ and (*b*) $M_b = 0.18$.

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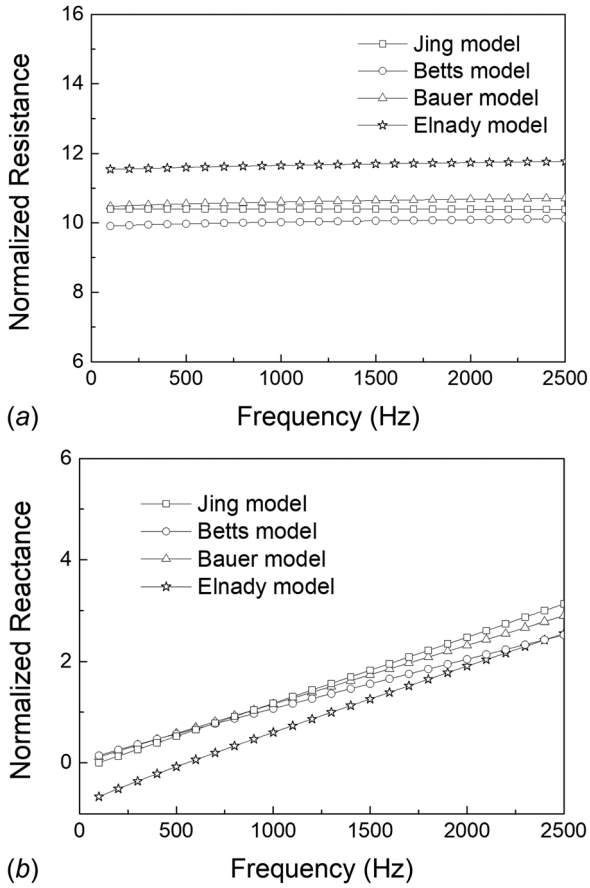

Fig. 10 Normalized impedance calculation results as a function of frequency according to impedance models ($M_b = 0.18$, $M_g = 0.06$, $\sigma = 0.02$, and SPL = 100 dB). Models of Jing, Betts, Bauer, and Elnady are from Eqs. (4), (5), (6), and (7)–(9), respectively: (*a*) normalized resistance and (*b*) normalized reactance

approximately constant over a frequency, whereas the reactance linearly increases with the frequency. The models of Jing, Betts, and Bauer show relatively equal resistance and reactance to each other. However, the Elnady model predicts a higher resistance and lower reactance than the others do. This high resistance of the Elnady model is mainly contributed by the effect of the bias flow term in Eq. (8). In this regard, it is noteworthy to compare the bias flow term in the Elnady model, "$(1.15M_b)/(C_d\sigma)$," with "$(4M_b)/(3\pi C_d^2\sigma)$" in Bauer model in Eq. (6). In addition, the grazing flow term of the Elnady model, "$-0.3M_g/\sigma$," in Eq. (9) results in a low reactance, whereas the other three models do not consider the grazing flow effect in the reactance term.

Figure 11 shows the effects of M_b and σ on the DC as a function of frequency with Jing's impedance model. The other impedance models predict a similar behavior with regard to M_b and σ. As shown in Fig. 11(*a*), the increase in M_b leads to the improvement of the low-frequency acoustic damping under approximately 800 Hz at a fixed σ of 0.02. However, the greatest damping is obtained at M_b of 0.06 over a broadband frequency range higher than 1000 Hz. Figure 11(*b*) plots the effect of σ at a given bias flow Mach number of 0.16. Overall, the maximum broadband acoustic damping exists approximately at a porosity of 0.03–0.05, except at frequencies lower than 500 Hz.

It would be interesting to analyze the conditions where the maximum dissipation can be obtained over various bias flow and porosity cases. From the previous literature [3,10,22,40], it is generally known that the ratio of the bias flow Mach number to porosity can be used as one of key parameters in determining the damping characteristics of perforated liners. Figure 12 plots the dissipation coefficients as a function of the ratio, M_b/σ, over the ranges of M_b and σ for three different frequencies. When plotting the results as a function of M_b/σ, all data from different M_b and σ conditions roughly collapse on a single curve for a given frequency. We also found that the optimal ratio, $(M_b/\sigma)_{\text{opt}}$, where the maximum dissipation occurs, decreases with increasing frequency.

Scarpato et al. [40] have analytically derived from Howe's model in Eqs. (2) and (3) that the $(M_b/\sigma)_{\text{opt}}$ lies within the range from 0.5 to $2/\pi$ depending on the Strouhal number of bias flow. Further, Eldredge and Dowling [22] suggested that $(M_b/\sigma)_{\text{opt}}$ of the acoustic liner should obey

$$\left(\frac{M_b}{\sigma}\right)_{\text{opt}} = \frac{A_{\text{surf}}/A_{\text{duct}}}{2\sqrt{2}} \tag{28}$$

where A_{surf} and A_{duct} mean the perforated surface area and the duct or combustor sectional area, respectively. In the current target combustor geometry as specified in Fig. 7 and Table 2, $A_{\text{surf}}/A_{\text{duct}}/2\sqrt{2}$ is calculated to be 0.47. As shown in Fig. 12, our calculated $(M_b/\sigma)_{\text{opt}}$ deviates significantly from the criteria of both Scarpato's and Eldredge's (i.e., 5.3 of $(M_b/\sigma)_{\text{opt}}$ for 700 Hz and 2.9 for 1700 Hz). A similar result is reported by Macquisten et al. [41] and Lahiri and Bake [10]. In particular, Lahiri and Bake [10] found that Eq. (28) can be used in choosing the maximum acoustic damping conditions for a relatively low bias flow Mach number where the resonance effect of liner back cavity dominates the damping mechanism. However, they reported that for high bias flow velocity conditions similar to the current analysis case, where no obvious resonance effect (such as a sharp peak absorption around the resonance frequencies) can be observed and, instead, the typical broadband damping is found as previously shown, the maximum absorption is obtained at a much higher M_b/σ than the value calculated in Eq. (28).

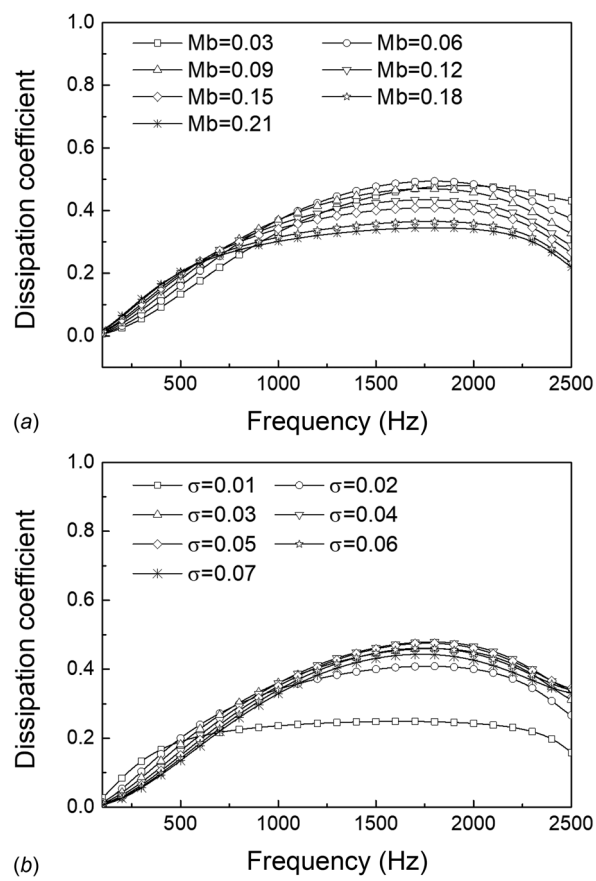

Fig. 11 Dissipation coefficient calculation results as a function of frequency using Jing's model in Eq. (4) ($M_g = 0.06$ and SPL = 100 dB): (*a*) effect of bias flow Mach number ($\sigma = 0.02$) and (*b*) effect of porosity ($M_b = 0.16$)

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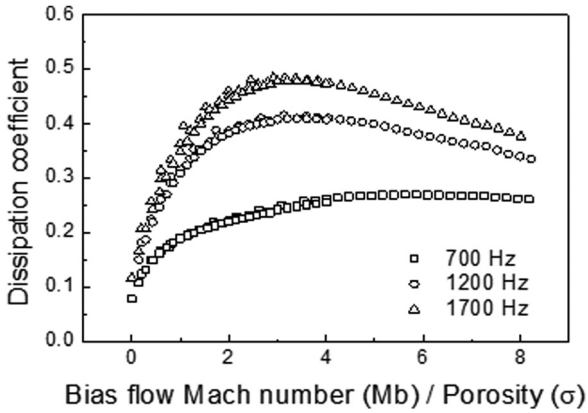

Fig. 12 Dissipation coefficient calculation results as a function of the ratio of bias flow Mach number to porosity using Jing's model in Eq. (4) ($M_b = 0.03$–0.21, $\sigma = 0.02, 0.05, 0.07$, $M_g = 0.06$, and SPL = 100 dB)

Figure 13 shows the DI defined in Eq. (25) as a function of the bias flow Mach number and porosity with four different frequency ranges under consideration using the Jing model. DI can be a more appropriate parameter in evaluating the overall broadband damping performance of perforated liners over frequencies than the DC [37]. As shown in the plot, similar to the dissipation coefficient, DI of the acoustic liner is highly dependent upon both M_b and σ. In Fig. 14, DI is replotted as a function of M_b/σ. Regardless of the change in M_b and σ, DI is merged onto the single characteristic curve in terms of the variable, M_b/σ, depending on the frequency range under consideration in the plot as already found in the DC of Fig. 12. The optimal M_b/σ lies in a relatively wide range for the case of DI defined between 50 and 1000 Hz, as shown in Fig. 14(*a*), whereas the relatively sharp peak is found at M_b/σ of approximately 2.9 for the case of DI defined between 50 and 2000 Hz, as shown in Fig. 14(*b*). DI defined between 50 and 2000 Hz appears to be a rational choice for the combustor under development, considering various modes of thermoacoustic combustion instability in a typical gas turbine combustor, such as bulk, longitudinal, and circumferential modes and their harmonic frequencies [1]. Once an optimal value of $(M_b/\sigma)_{\text{opt}}$ for the maximum acoustic energy absorption is provided, the optimal combination of the number (n) and diameter (d) of perforation orifices can be obtained from Eqs. (26) and (27) for a given bias (or liner cooling) flowrate ($\dot{m}_b$) as follows:

$$(n \cdot d^2)_{\text{opt}} = 4\sqrt{\frac{\dot{m}_b D_1 l_{\text{tot}}}{\pi \rho c (M_b/\sigma)_{\text{opt}}}} \tag{29}$$

Finally, Fig. 15 shows the suggested number of holes at a given hole diameter such that the maximum absorption through the perforated liner can be achievable over frequencies of 50–2000 Hz using Eq. (29). Under this condition, the optimal values of M_b/σ obtained by the models of Betts, Bauer, and Elnady are calculated as 3.0, 3.1, and 3.3, respectively. As seen in the plot, the deviation of the suggested number of holes between the models is not remarkable. As an example, the equivalent porosities calculated using the suggested number of holes are 0.101 by the Elnady model and 0.107 by the Jing model, respectively.

The current design basis of a perforated liner is presented only from the viewpoint of acoustic damping. It needs to be further optimized through the consideration of other design disciplines of the combustor liner, such as heat transfer and emission in our

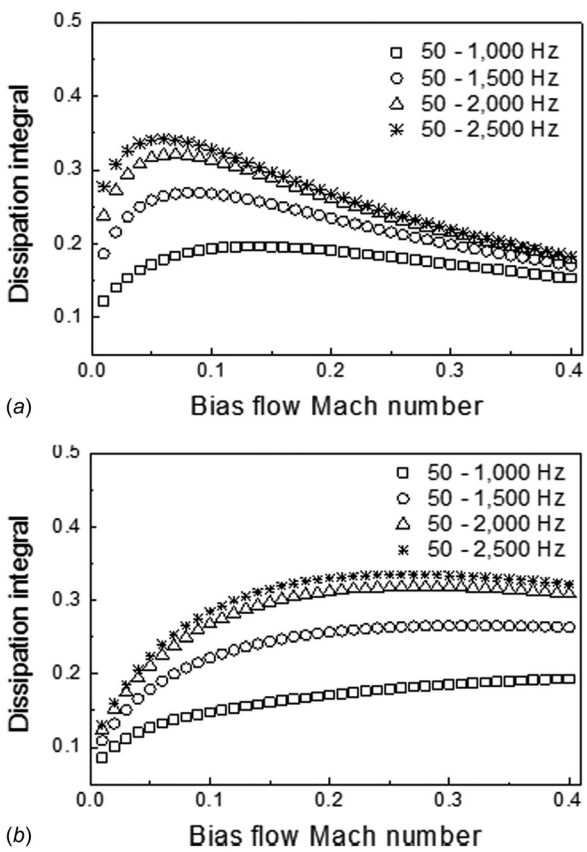

Fig. 13 Dissipation integral calculation results as a function of bias flow Mach number using Jing's model in Eq. (4) ($M_g = 0.06$ and SPL = 100 dB): (*a*) σ=0.025 and (*b*) $\sigma = 0.08$

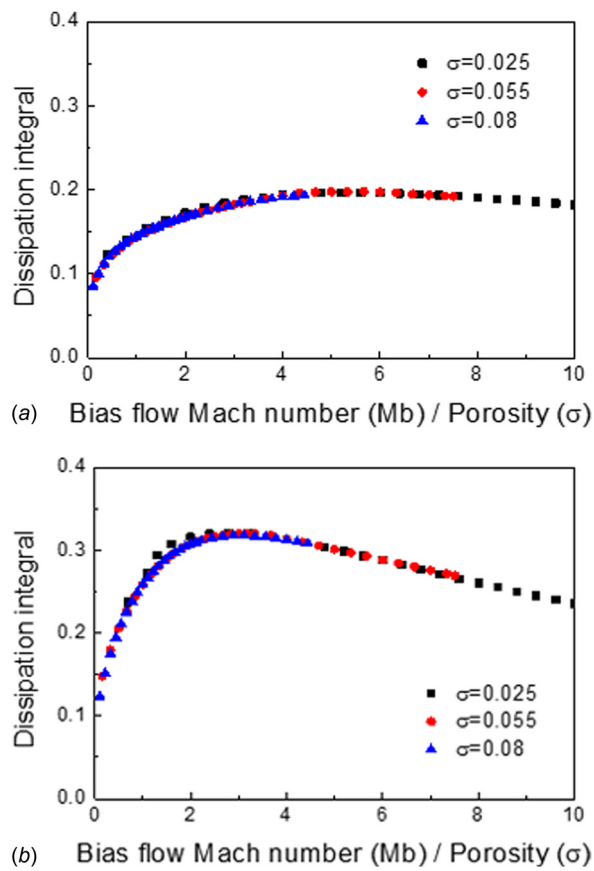

Fig. 14 Dissipation integral calculation results as a function of the ratio of bias flow Mach number to porosity using Jing's model in Eq. (4) ($M_g = 0.06$ and SPL = 100 dB): (*a*) DI (50–1000 Hz) and (*b*) DI (50–2000 Hz)

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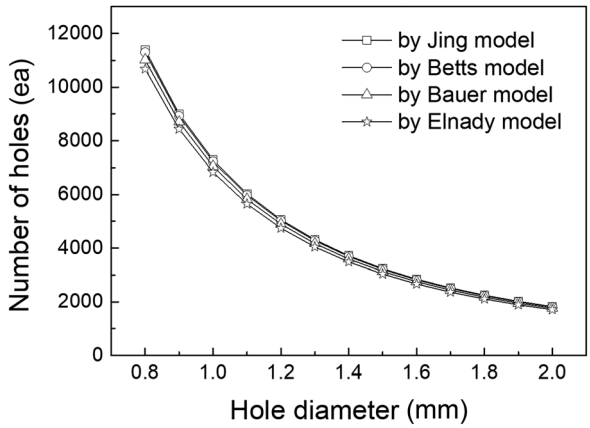

Fig. 15 Suggested number of holes at a given hole diameter using Eq. (29)

ongoing development processes. In addition, the present analysis was simplified by assuming that the pressure fluctuations are propagated by one-dimensional plane waves in the combustor and cavities. This is not strictly true especially for an annular combustor, where higher acoustic modes in the circumferential and radial directions can be cut on. The current model will be extended to account for the higher acoustic mode in a perforated liner using three-dimensional approaches [42–44] as future work.

Conclusions

Four popular zero-dimensional impedance models, such as the models of Jing, Bauer, Betts, and Elnady for a perforated plate are incorporated into the transfer matrix method in order to analyze the acoustic damping in a two-duct system with a perforated element. Thereafter, the models are validated against experimental data from the literature under high bias flow velocity conditions. The validation results show that all of the impedance models considered in this study can appropriately represent the quantitative effects of the bias flow Mach number and porosity of the perforated liner against the measurement. The impedance models are then applied to a small aircraft gas turbine combustor under development to find an optimal liner geometry in terms of the maximum acoustic energy absorption at a given design combustor configuration and operating condition. The Elnady model predicts slightly lower damping characteristics under a given condition than the other impedance models do. However, this deviation is negligibly small. Moreover, it is found that the ratio of the bias flow Mach number to the porosity can be used as a design guideline in choosing the optimal combination of the number and diameter of perforations in terms of the acoustic damping. The current finding will be further optimized through the consideration of other design disciplines of the combustor liner, such as heat transfer and emission in our ongoing development processes.

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Nomenclature

A = area
c = speed of sound
C_d = orifice discharge coefficient
d = orifice diameter
D = duct diameter
DC = dissipation coefficient
DI = dissipation integral
f = frequency
h = cavity height
I_1 = modified Bessel function of the first kind
J_0 = zeroth-order Bessel function of the first kind
J_1 = first-order Bessel function of the first kind
k = wavenumber
K_R = Rayleigh conductivity
K_s = Stokes wavenumber
K_1 = modified Bessel function of the second kind
l = length
M = Mach number
$\dot{m}$ = mass flowrate
n = number of orifice
p = pressure
p_{ref} = reference pressure
RC = reflection coefficient
St = Strouhal number
t = wall thickness
T = temperature
$\mathbf{T}$ = transfer matrix
TC = transmission coefficient
u = velocity
x = axial position
z = normalized acoustic impedance

Greek Symbols

γ = real part of Rayleigh conductivity
δ = imaginary part of Rayleigh conductivity
Θ = normalized resistance
ν = kinematic viscosity
ρ = density
σ = porosity
X = normalized reactance
ψ = interaction factor
ω = angular frequency

Subscripts

b = bias flow
down = downstream
end = end correction
g = grazing flow
max = maximum
min = minimum
surf = liner surface
tot = total
up = upstream
$1, 2$ = duct index

Superscripts

$+$ = positive-going wave
$-$ = negative-going wave
$\bar{\ }$ = mean value
$\hat{\ }$ = complex amplitude

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